

MLCA

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```
## -----  
##' [libraries]  
## -----
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.0      v readr      2.1.5  
v forcats    1.0.1      v stringr    1.5.2  
v ggplot2    4.0.0      v tibble     3.3.0  
v lubridate  1.9.4      v tidyr      1.3.1  
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

```
library(tidycensus)
```

```
library(multilevLCA)
```

multilevLCA: Estimates and plots single- and multilevel latent class models.

```
library(psych)
```

Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
library(tigris)
```

To enable caching of data, set ``options(tigris_use_cache = TRUE)`` in your R script or `.Rprofile`.

```
library(nnet)  
library(VGAM)
```

Loading required package: stats4
Loading required package: splines

Attaching package: 'VGAM'

The following object is masked from 'package:psych':

logistic

```
library(gt)  
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:psych':

logit

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

```
rm(list = ls())
```

```
## -----
##' [IPEDS/Carnegie Classification Data]
## -----
```

```
# load IPEDS data and merge with the Carnegie Classification
```

```
ipedes24 <- read_csv("../data/sources/hd2024.csv")
```

```
Rows: 6072 Columns: 72
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (23): INSTNM, IALIAS, ADDR, CITY, STABBR, ZIP, CHFM, CHFTITLE, EIN, UEI...
```

```
dbl (49): UNITID, FIPS, OBEREG, GENTLE, OPEFLAG, SECTOR, ICLEVEL, CONTROL, ...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ipedes_cost <- read_csv("../data/sources/cost1_2024.csv")
```

```
Rows: 5867 Columns: 418
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (198): XROOMCAP, XMEALSWK, XAPPFEEU, XAPPFEEG, XAPPFEED, FAFQ2URL, XROOM...
```

```
dbl (214): UNITID, ROOM, ROOMCAP, ALLONCAM, TUITVARY, BOARD, MEALSWK, APPLFE...
```

```
lgl (6): CHG10AY0, CHG10AY1, CHG10AY2, CHG10PY0, CHG10PY1, CHG10PY2
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#ipedes_schoolsize <- read_csv("../data/sources/ef2024a.csv") #enrollment Fall 2024
carnegie <- read_excel("../data/sources/2025-Public-Data-File.xlsx", sheet = 2)
ic2024 <- read_csv("../data/sources/ic2024.csv")
```

```
Rows: 5963 Columns: 111
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): XDISABPC
```

```
dbl (110): UNITID, PE01ISTR, PE02ISTR, PE03ISTR, PE04ISTR, PE05ISTR, PE06IST...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
sfa <- read_csv("../data/sources/sfa2324.csv")
```

Rows: 5575 Columns: 183

-- Column specification -----

Delimiter: ","

chr (91): XSCUGRAD, XSCUGDGSK, XSCUGNDGS, XSCUGFFN, XSCUGFFP, XSCFA2, XSCFA2...

dbl (92): UNITID, SCUGRAD, SCUGDGSK, SCUGNDGS, SCUGFFN, SCUGFFP, SCFA2, SCFA...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
adm <- read_csv("../data/sources/adm2024.csv")
```

Rows: 1956 Columns: 101

-- Column specification -----

Delimiter: ","

chr (44): XAPPLCN, XAPPLCNM, XAPPLCNW, XAPPLCNAN, XAPPLCNUN, XADMSSN, XADMSS...

dbl (57): UNITID, ADMCON1, ADMCON2, ADMCON3, ADMCON4, ADMCON5, ADMCON6, ADMC...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
enroll <- read_csv("../data/sources/effy2024.csv") %>%
  filter(EFFYALEV == 3) %>% #degree/certificate-seeking undergraduates
  select(UNITID, EFFYALEV, EFYTOTLT, EFYAIANT, EFYBKAAT, EFYNHPIT, EFYWHITT, EFYHISPT, EFYAS...
```

Rows: 115026 Columns: 72

-- Column specification -----

Delimiter: ","

chr (34): XEYTOTLT, XEYTOTLM, XEYTOTLW, XEFYAIAT, XEFYAIAM, XEFYIAIW, XEFYAS...

dbl (38): UNITID, EFFYALEV, EFFYLEV, LSTUDY, EFYTOTLT, EFYTOTLM, EFYTOTLW, E...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
#ipeds_distance <- read_csv("../data/sources/effy2024_dist.csv") %>%
# filter(EFFYDLEV == 2) %>% #undergrad total
# select(UNITID, EFFYDLEV, EFYDETOT, EFYDEEXC)
ipeds_dist <- read_csv("../data/sources/ef2024a_dist.csv") |>
  filter(EFDELEV == 2) |> #undergrad total
  select(UNITID, EFDELEV, EFDETOT, EFDEEXC, XEFDEEXC)
```

Rows: 21819 Columns: 20

-- Column specification -----

Delimiter: ","

chr (9): XEFDETOT, XEFDEEXC, XEFDESOM, XEFDENON, XEFDEEX1, XEFDEEX2, XEFDEE...

dbl (11): UNITID, EFDELEV, EFDETOT, EFDEEXC, EFDESOM, EFDENON, EFDEEX1, EFDE...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
ipeds_learner <- read_csv("../data/sources/ef2024b.csv") |>
  filter(LSTUDY == 2, EFBAGE %in% c(1, 7)) |>      #undergrad total / all age categories total
#number of institutions (= UNITID): 2,487
  select(UNITID, EFBAGE, EFAGE06, EFAGE09) |>
  pivot_wider(names_from = EFBAGE,
              values_from = c(EFAGE06, EFAGE09))
```

Rows: 76359 Columns: 22

-- Column specification -----

Delimiter: ","

chr (9): XEFAGE01, XEFAGE02, XEFAGE03, XEFAGE04, XEFAGE05, XEFAGE06, XEFAGE...

dbl (13): UNITID, EFBAGE, LINE, LSTUDY, EFAGE01, EFAGE02, EFAGE03, EFAGE04, ...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
#Let's replace this with the size variable in IPEDS (instsize)
# ipeds_schoolsize <- ipeds_schoolsize |>
#   filter(EFALEVEL == 1) |> #Level of student (1 == all students total)
#   select(UNITID, EFTOTLT) |> #Grand total
#   mutate(size = case_when(
#     EFTOTLT <= 3000 ~ 0, #small
#     EFTOTLT <= 10000 ~ 1, #medium
#     EFTOTLT > 10000 ~ 2, #large
#     TRUE ~ NA_real_))

carnegie <- carnegie |>
  select(unitid, hsi)

# merge files
data <- ipeds24 %>%
  left_join(carnegie, by = c("UNITID" = "unitid")) %>%
  left_join(ipeds_cost, by = "UNITID") |>
```

```

#left_join(ipeds_schoolsize, by = "UNITID") %>%
left_join(ic2024, by = "UNITID" ) %>%
left_join(sfa, by = "UNITID") %>%
left_join(adm, by = "UNITID") %>%
left_join(enroll, by = "UNITID") |>
left_join(ipeds_dist, by = "UNITID") |>
left_join(ipeds_learner, by = "UNITID")

# select variables and create a dataset for analysis
x <- data |>
  rename_all(tolower) |> #converts all column names in a data frame to lowercase
  select(unitid, instnm, city, stabbr, fips, countycd, countynm, cbsa, cbsatype, csa, ober
    locale, zip, longitud, latitude,
    control, iclevel, instcat, deggrant, instsize,
    hloffer,
    carnegieic, carnegiesaec, carnegiealf, carnegieapm, carnegiersch,
    hbcu, hsi, tribal,
    room, relaffil, distnced, athassoc,
    admsn, applcn, openadmp, loan_a, tuition1, upgrntp,
    efytotlt, efyaiant, efybkaat, efyhnpit, efywhitt, efyhispt, efyasiat,
    efdelev, efdetot, efdeexc, xefdeexc, slo7, eface06_1, eface06_7, eface09_1, eface
# assign to "data"
data <- x

```

```

x <- data %>%
  filter(fips <= 56) |> # Keep only 50 states and DC
  filter(cbsa != -2) |> # Remove colleges not located in CBSA
  # filter(!is.na(control), control != 3) |> # 3: private for-profit
  filter(!is.na(iclevel), iclevel != 3) |> # 3: less than 2 years
  filter(!is.na(carnegiealf), carnegiealf != 6) %>% # 6: graduate-focused
  filter(instcat >= 2 & instcat <= 4) # 1: degree-granting, graduate with 1
  #filter(carnegieapm == 1) # 5-6: non-degree-granting
  # Keep only colleges classified as "1"

data <- x

```

```

## -----
##' [Data Cleaning]
## -----

x <- data |>
  mutate(
    public = case_when(
      control == 2 ~ 0,          # 0: private
      control == 1 ~ 1,          # 1: public
      control == 3 ~ 2,          # 2: for-profit
      TRUE ~ NA_real_),

    twoyear = case_when(
      iclevel == 1 ~ 0,          # 0: 4 or more
      iclevel == 2 ~ 1,          # 1: at least 2 & less than 4
      TRUE ~ NA_real_),

    # msi = case_when(
    #   hbcu == 1 ~ 1,            # 0: non of them
    #   hsi == 1 ~ 2,            # 1: HBCU
    #   tribal == 1 ~ 3,         # 2: HSI
    #   hbcu == 2 & hsi == 0 & tribal == 2 ~ 0,
    #   is.na(hbcu) | is.na(hsi) | is.na(tribal) ~ NA_real_,
    #   TRUE ~ NA_real_),
    #   TRUE ~ 0),

    rurality = case_when(
      locale %in% c(31,32,33,41,42,43) ~ 0,    # 0: Town + Rural
      locale %in% c(21,22,23) ~ 1,             # 1: Suburb
      locale %in% c(11,12,13) ~ 2,             # 2: City
      TRUE ~ NA_real_),

    size = case_when(
      instsize == 1 ~ 0,          # 0: < 1,000
      instsize == 2 ~ 1,          # 1: 1,000 - 4,999
      instsize == 3 ~ 2,          # 1: 5,000 - 9,999
      instsize == 4 ~ 3,          # 2: 10,000 - 19,999
      instsize == 5 ~ 4,          # 3: >=20,000
      TRUE ~ NA_real_),
  )

```

```

#       tuition = case_when(
#         tuition1 == 0 & obereg != 0 ~ NA_real_,
#         tuition1 == 0 & obereg == 0 ~ 0,
#         tuition1 < 10000 ~ 0,           # 0: < $10,000           - 1334
#         tuition1 >= 10000 & tuition1 < 20000 ~ 1,       # 1: $10,000 - $19,999 - 689
#         # tuition1 >= 20000 & tuition1 < 30000 ~ 2,     # 2: $20,000 - $29,999
#         tuition1 >= 20000 & tuition1 < 40000 ~ 2,       # 3: $30,000 - $39,999 - 531
#         # tuition1 >= 40000 & tuition1 < 50000 ~ 4,     # 4: $40,000 - $49,999
#         tuition1 >= 40000 ~ 3,           # 5: > $50,000           - 384
#         TRUE ~ NA_real_),              #                      - 316

#       tuitionnn = ntile(tuition1, 4) - 1,              # 0~3: quartiles

#check missing values: sum(is.na(ipeds_cost$TUITION1)) [1] 2,368 (= about 40% of the sample)

housing = case_when(
  room == 2 ~ 0,           # 0: no housing
  room == 1 ~ 1,           # 1: offer housing
  TRUE ~ NA_real_),

religious = ifelse(relaffil == -2, 0, 1),               # 0: no religious
                                                         # 1: any religious

#We need to remove colleges that offer completely distance education from our sample
distance = case_when(
  distnced == 2 ~ 0,       # 0: not completely distanced learning
  distnced == 1 ~ 1,       # 1: completely distanced learning
  TRUE ~ NA_real_),

online_perc = case_when(
  is.na(efdetot) ~ NA_real_,
  is.na(efdeexc) & xefdeexc == "A" ~ 0,               # "A": not applicable
  is.na(efdeexc) ~ NA_real_,                          # other missing cases
  TRUE ~ efdeexc / efdetot),

#       online = case_when(
#         is.na(online_perc) ~ NA_real_,
#         online_perc == 0 ~ 0,           # none - 8
#         online_perc < 0.25 ~ 1,        # more than 0%, less than 25% - 1
#         online_perc < 0.50 ~ 2,        # 25% to less than 50% - 7
#         online_perc <= 1 ~ 3,          # 50% to 100% - 2
#         TRUE ~ NA_real_),              # NA - 1

```

```

    online = ntile(online_perc, 4) - 1,          # 0~3: quartiles

#check missing values: sum(is.na(ipeds_dist$EFDETOT))  [1] 0

    evening = case_when(
      slo7 == 0 ~ 0,          # 0: no weekend/evening courses
      slo7 == 1 ~ 1,          # 1: allows classes only on weekend
      TRUE ~ NA_real_),

#check missing values: sum(is.na(ic2024$SL07))  [1] 0

#      adult_perc = case_when(
#        is.na(efage09_7) ~ NA_real_,
#        is.na(efage09_1) ~ NA_real_,
#        TRUE ~ efage09_7 / efage09_1),

#      adult_perc = case_when(
#        is.na(efage09_7) ~ NA_real_,
#        is.na(efage09_1) ~ NA_real_,
#        TRUE ~ efage09_7 / efage09_1),

#      adult = ntile(adult_perc, 4) - 1,          # 0~3: quartiles

#check missing values: sum(is.na(ipeds_learner$EFAGE09_1))  [1] 0
#check missing values: sum(is.na(ipeds_learner$EFAGE09_7))  [1] 9

#      parttime_perc = case_when(
#        is.na(efage06_1) ~ NA_real_,
#        is.na(efage09_1) ~ NA_real_,
#        TRUE ~ efage06_1 / efage09_1),

#      parttime = case_when(
#        is.na(parttime_perc) ~ NA_real_,
#        parttime_perc == 0 ~ 0,          # none - 7%
#        parttime_perc < 0.25 ~ 1,        # more than 0%, less than 25% - 8%
#        parttime_perc < 0.50 ~ 2,        # 25% to less than 50% - 34%
#        parttime_perc <= 1 ~ 3,          # 50% to 100% - 70%
#        TRUE ~ NA_real_),              # NA - 14%

#      parttime = ntile(parttime_perc, 4) - 1,    # 0~3: quartiles
#check missing values: sum(is.na(ipeds_learner$EFAGE06_1))  [1] 0

```

```

sports = case_when(
  athassoc == 2 ~ 0,          # 0: not a member
  athassoc == 1 ~ 1,          # 1: member of National Athletic Assn
  TRUE ~ NA_real_),

loan = case_when(
  is.na(loan_a) ~ NA_real_,
  loan_a <= quantile(loan_a, 0.25, na.rm = TRUE) ~ 0,    # 0: lowest
  loan_a <= quantile(loan_a, 0.50, na.rm = TRUE) ~ 1,    # 1: low
  loan_a <= quantile(loan_a, 0.75, na.rm = TRUE) ~ 2,    # 2: high
  TRUE ~ 3),        # 3: highest

```

Q. Since MLCA requires variables starting at 0, is it appropriate to divide loan amount into 4 categories?

```

asc = case_when(
  carnegiesaec == 2 ~ 0,      # 0: Higher Access, Lower Earnings
  carnegiesaec == 4 ~ 1,      # 1: Higher Access, Medium Earnings
  carnegiesaec == 6 ~ 2,      # 2: Higher Access, Higher Earnings
  carnegiesaec == 1 ~ 3,      # 3: Lower Access, Lower Earnings
  carnegiesaec == 3 ~ 4,      # 4: Lower Access, Medium Earnings
  carnegiesaec == 5 ~ 5,      # 5: Lower Access, Higher Earnings
  TRUE ~ NA_real_),

awardlev = case_when(
  carnegiealf %in% c(1,2) ~ 0, # 0: associates
  carnegiealf == 3 ~ 1,        # 1: baccalaureate
  carnegiealf == 4 ~ 2,        # 2: undergrad / grad-Master's
  carnegiealf == 5 ~ 3,        # 3: undergrad / grad-Doctorate

  is.na(carnegiealf) & (hloffer == 3 | hloffer == 4) ~ 0,
  carnegiealf == -2 & (hloffer == 3 | hloffer == 4) ~ 0,
  hloffer == 5 & instcat == 2 ~ 1,

  TRUE ~ NA_real_),
#Check: x %>% filter(is.na(awardlev)) %>% View()

focus = case_when(
  carnegieapm == -2 ~ 0, # 0: Not classified
  carnegieapm == 1 ~ 1, # 1: Mixed
  carnegieapm == 2 ~ 2, # 2: Professions-focused
  carnegieapm == 3 ~ 3, # 3: Applied and career studies
  carnegieapm %in% c(4,5) ~ 4, # 4: Arts and sciences

```

```

    carnegieapm == 6 ~ 5, # 5: Business
    carnegieapm %in% c(8,9,10) ~ 6, # 6: Medical sciences/health
    carnegieapm == 11 ~ 7, # 7: STEM
    carnegieapm == 12 ~ 8, # 8: Theology
    TRUE ~ NA_real_),

research = case_when(
  carnegiersch == -2 ~ 0, #0: No research designation
  carnegiersch == 3 ~ 1, #1: Research Colleges and Universities
  carnegiersch == 2 ~ 2, #2: High Spending
  carnegiersch == 1 ~ 3, #3: Very High Spending
  TRUE ~ NA_real_),

cbsa = as.character(cbsa),
cbsatype = as.character(cbsatype),

pell = case_when(
  upgrntp < 10 ~ 0,
  upgrntp >= 10 & upgrntp < 30 ~ 1,
  #upgrntp >= 20 & upgrntp < 30 ~ 2,
  upgrntp >= 30 & upgrntp < 50 ~ 2,
  # upgrntp >= 40 & upgrntp < 50 ~ 4,
  upgrntp >= 50 ~ 3,
  TRUE ~ NA_real_)

)

data <- x |>
  filter(distance == 0) %>% # remove observations with only distance e
  group_by(cbsa) %>%
  mutate(inst_num = n_distinct(unitid),
         inst_num = as.numeric(inst_num)) %>%
  ungroup() %>%
  mutate(admrate = admssn/applcn) %>%
  # mutate(admrate = if_else(openadmp == 1, 1, admrate)) |>
  mutate(selectivity = case_when(
    openadmp == 1 ~ 0,
    admrate == 1 ~ 0, #0: open
    admrate >= 0.75 & admrate < 1 ~ 1, #1: less selective
    admrate >= 0.40 & admrate < 0.75 ~ 2, #2: selective
    admrate < 0.40 ~ 3, #3: more selective
  ))

```

```

    TRUE ~ NA_real_),

    i_perc_black = efybkaat/efytotlt,           #Percent of degree-seeking undergrad
    i_perc_white = efywhitt/efytotlt,
    i_perc_hisp = efyhispt/efytotlt,
    i_perc_nhpi = (efyaiant + efynhpi)/efytotlt,
    i_perc_asian = efyasiat/efytotlt,
    i_perc_nonwhite = 1 - i_perc_white,
    nonwhite = ntile(i_perc_nonwhite, 4),
    nonwhite = nonwhite - 1
  ) %>%
rename(region = obereg) %>%
mutate(cbsa = as.character(cbsa))

```

```

z <- read_csv("../data/sources/effy2024.csv") %>%
  filter(EFFYLEV == 2 | EFFYLEV == 4) %>%
  select(UNITID, EFFYALEV, EFTOTLT) %>%
  mutate(EFFYALEV = if_else(EFFYALEV == 2, "undergrad", "grad")) %>%
  pivot_wider(id_cols = "UNITID",
              names_from = EFFYALEV,
              values_from = EFTOTLT) %>%
  mutate(total_pop = undergrad + grad,
         per_undergrad = undergrad/total_pop,
         per_grad = grad/total_pop)

```

Rows: 115026 Columns: 72

-- Column specification -----

Delimiter: ","

chr (34): XEYTOTLT, XEYTOTLM, XEYTOTLW, XEFYAIAT, XEFYAIAM, XEFYAIAM, XEFYAS...

dbl (38): UNITID, EFFYALEV, EFFYLEV, LSTUDY, EFTOTLT, EFTOTLM, EFTOTLW, E...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

## -----
## [Census/ACS Data]
## -----

land_area <- core_based_statistical_areas(
  cb = TRUE,
  year = 2024) %>%
  select(GEOID, ALAND)

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=====	98%
=====	99%
=====	99%
=====	100%

```
census <- get_acs(geography = "cbsa",
  year = 2024,
  variables = c("B01003_001E", # Total population
    "B19013_001E", # Median Household Income in the Past 12 Months
    "DP03_0119PE", # % families below poverty line
    "DP03_0005PE", # % unemployed over 16 years old
```

```

        "DP02_0068PE", # % bach or higher
        "B03002_003E", # total white (alone, not hispanic)
        "B03002_004E", # total black (alone, not hispanic)
        "B03002_012E", # total hispanic
        "B03002_006E", # total asian (alone, not hispanic)
        "B03002_001E" # total by ethnicity
    ),
    output = "wide") %>%
left_join(land_area)

```

Getting data from the 2020-2024 5-year ACS

Fetching data by table type ("B/C", "S", "DP") and combining the result.

Joining with `by = join\_by(GEOID)`

```

census <- census |>
  mutate(perc_white = B03002_003E/B03002_001E*100,
         perc_black = B03002_004E/B03002_001E*100,
         perc_hisp = B03002_012E/B03002_001E*100,
         perc_asian = B03002_006E/B03002_001E*100,
         tot_pop = B01003_001E/100000,
         pop_density = B01003_001E/(ALAND/2589988.10)) |> #population per sq mile
  rename(perc_ba = DP02_0068PE,
         med_income = B19013_001E,
         perc_pov = DP03_0119PE,
         perc_unemployed = DP03_0005PE) |>
  select(tot_pop, pop_density, perc_ba,
         med_income, perc_pov, perc_unemployed,
         perc_white, perc_black, perc_hisp, perc_asian,
         GEOID) |>
  mutate(GEOID = as.character(GEOID))

data <- data |>
  left_join(census, by = c("cbsa" = "GEOID"))

```

```

data_mlca <- data %>%
  select(unitid, instnm, stabbr, cbsa, cbsatype, region,
         public, twoyear, size, rurality,
         awardlev,

```

```

    religious, focus, housing, research, msi,
    inst_num, selectivity, pell, online, evening, #adult, tuitionnn, parttime,
    tot_pop, pop_density, perc_ba, med_income, perc_pov, perc_unemployed,
    perc_white, perc_black, perc_hisp, perc_asian,
    i_perc_white, i_perc_black, i_perc_hisp, i_perc_asian, nonwhite) %>%
as.data.frame()

#metropolitan, tuition,msi, distance,sports,research,

describe(data_mlca)

```

	vars	n	mean	sd	median	trimmed	mad
unitid	1	3254	238590.85	120539.08	198911.00	223531.74	59118.67
instnm*	2	3254	1610.98	932.84	1608.50	1610.43	1200.16
stabbr*	3	3254	25.69	14.59	26.00	25.72	19.27
cbsa*	4	3254	374.03	200.50	400.00	377.35	246.11
cbsatype*	5	3254	1.14	0.35	1.00	1.06	0.00
region	6	3254	4.49	2.12	5.00	4.44	2.97
public	7	3254	0.80	0.72	1.00	0.74	1.48
twoyear	8	3254	0.33	0.47	0.00	0.29	0.00
size	9	3254	1.20	1.18	1.00	1.05	1.48
rurality	10	3254	1.27	0.82	2.00	1.33	0.00
awardlev	11	3232	0.99	1.12	1.00	0.87	1.48
religious	12	3254	0.21	0.41	0.00	0.14	0.00
focus	13	3254	2.74	2.04	2.00	2.46	1.48
housing	14	3251	0.55	0.50	1.00	0.56	0.00
research	15	3254	0.30	0.79	0.00	0.06	0.00
msi	16	3254	0.18	0.38	0.00	0.10	0.00
inst_num	17	3254	35.52	56.68	13.00	20.58	16.31
selectivity	18	3165	0.78	0.92	0.00	0.66	0.00
pell	19	3240	1.87	0.81	2.00	1.86	1.48
online	20	3244	1.48	1.11	1.00	1.47	1.48
evening	21	3254	0.57	0.50	1.00	0.59	0.00
tot_pop	22	3254	36.27	53.61	10.38	23.39	14.56
pop_density	23	3254	743.48	897.54	359.84	544.81	390.87
perc_ba	24	3254	35.39	9.19	36.50	35.42	8.60
med_income	25	3254	81837.51	18132.35	79878.50	80879.19	17562.14
perc_pov	26	3254	8.87	2.87	8.60	8.56	2.22
perc_unemployed	27	3254	3.27	0.77	3.20	3.27	0.74
perc_white	28	3254	59.50	19.37	60.78	59.99	24.36
perc_black	29	3254	11.87	10.32	8.29	10.29	9.33
perc_hisp	30	3254	17.62	15.41	12.22	15.21	11.59

perc_asian	31	3254	5.51	5.95	3.21	4.46	3.13
i_perc_white	32	3248	0.47	0.25	0.50	0.48	0.30
i_perc_black	33	3248	0.17	0.20	0.10	0.13	0.10
i_perc_hisp	34	3248	0.18	0.18	0.12	0.15	0.11
i_perc_asian	35	3248	0.05	0.07	0.02	0.03	0.02
nonwhite	36	3248	1.50	1.12	1.50	1.50	1.48
		min	max	range	skew	kurtosis	se
unitid		100654.00	500333.00	399679.00	1.13	-0.18	2113.10
instnm*		1.00	3226.00	3225.00	0.00	-1.20	16.35
stabbr*		1.00	51.00	50.00	-0.09	-1.34	0.26
cbsa*		1.00	725.00	724.00	-0.20	-1.15	3.51
cbsatype*		1.00	2.00	1.00	2.02	2.09	0.01
region		0.00	8.00	8.00	0.18	-0.98	0.04
public		0.00	2.00	2.00	0.33	-1.05	0.01
twoyear		0.00	1.00	1.00	0.70	-1.50	0.01
size		0.00	4.00	4.00	0.89	-0.07	0.02
rurality		0.00	2.00	2.00	-0.53	-1.31	0.01
awardlev		0.00	3.00	3.00	0.72	-0.91	0.02
religious		0.00	1.00	1.00	1.43	0.03	0.01
focus		0.00	8.00	8.00	1.13	0.30	0.04
housing		0.00	1.00	1.00	-0.18	-1.97	0.01
research		0.00	3.00	3.00	2.61	5.50	0.01
msi		0.00	1.00	1.00	1.68	0.81	0.01
inst_num		1.00	217.00	216.00	2.33	4.46	0.99
selectivity		0.00	3.00	3.00	0.86	-0.36	0.02
pell		0.00	3.00	3.00	0.05	-1.08	0.01
online		0.00	3.00	3.00	0.02	-1.34	0.02
evening		0.00	1.00	1.00	-0.28	-1.92	0.01
tot_pop		0.13	197.99	197.86	1.97	3.02	0.94
pop_density		2.61	3224.72	3222.11	1.73	1.94	15.73
perc_ba		11.40	69.70	58.30	-0.07	-0.37	0.16
med_income		31354.00	162111.00	130757.00	0.75	1.42	317.87
perc_pov		2.80	30.50	27.70	1.76	6.42	0.05
perc_unemployed		0.40	6.60	6.20	0.13	-0.01	0.01
perc_white		3.58	94.47	90.89	-0.20	-0.86	0.34
perc_black		0.05	74.45	74.40	1.53	3.02	0.18
perc_hisp		0.48	95.00	94.52	1.46	2.24	0.27
perc_asian		0.01	41.61	41.60	2.63	10.01	0.10
i_perc_white		0.00	1.00	1.00	-0.09	-0.93	0.00
i_perc_black		0.00	1.00	1.00	2.01	3.80	0.00
i_perc_hisp		0.00	1.00	1.00	1.74	2.99	0.00
i_perc_asian		0.00	0.95	0.95	3.64	22.04	0.00
nonwhite		0.00	3.00	3.00	0.00	-1.36	0.02

```

x <- data_mlca %>%
  drop_na() %>%
  group_by(cbsa) %>%
  mutate(
    cbsa_id = cur_group_id()
  ) %>%
  as.data.frame()

# data_mlca %>% anti_join(x, by = "unitid") %>% View()

data_mlca <- x

write_rds(data_mlca, "../data/analytic_v2")

# Variables:
#   Y-          Columns with indicator variables for level 1 class membership (predictors of i
#               # public, two_year, size, msi, award level, research, selectivity

#   iT-         Number of lower-level latent classes

#   id_high-    Name of column with higher-level id
#               # CBSA

#   iM-         Number of higher-level latent classes

#   Z-          Columns with lower-level covariates (predictors of level 1 class membership)
#               # rurality, housing, religious, sports

#   Zh-         Columns with higher-level covariates (predictors of level 2 class membership)
#               # Variables- region, number of IHEs, total population, percent bachelors deg

# Level-1 indicators: Observed institution characteristics used to define the latent institut
# Level-1 covariates: Institution-level predictors that explain class membership
# Level-2 covariates: CBSA-level regional variables that explain differences in institution c

# Lower-level indicators based on framework for insitutional diversity in:
# Harris, M.S. An Empirical Typology of the Institutional Diversity of U.S. Colleges and Uni

#indicators
Y <- c(

```

```

#systemic- structural and governance characteristics
"public", "size", #"twoyear",

#programmatic- degree level and academic disciplinary offerings
"awardlev", "focus", "research",

#procedural- how programs are offered and implemented (Harris included % of students enrol.
#-the institution / management? operation?
"online", #ADD VARIABLE FOR THE PERCENTAGE OF STUDENTS WHO ARE EXCLUSIVELY ENROLLED IN ONLINE
#categorized share of undergraduates enrolled exclusively in distance education
"evening",
# "tuitionnn",
#(1)including tuition as quartiles: iT= 8 & iM= 2 (#obs = 1779)
#(2)including tuition & parttime quartiles: iT= 8 & iM= 2 (#obs = 1779)

#constituential- "student demographics, background, preparation, and abilities" (Harris, 2000)
"selectivity", "pell", "msi", "nonwhite" #, "adult", "parttime"
) #level-1 indicators

Z <- c("i_perc_white", "i_perc_black", "i_perc_hisp", "i_perc_asian", "housing", "rurality",
Zh <- c("cbsatype", "inst_num", "tot_pop", "pop_density", "perc_ba", "med_income", "perc_pov")

```

```

# Level-1 indicators (Y)
## cf. these are not predictors, so technically we don't need to check for multicollinearity
## cf. these are categorical variables, so VIF may not be appropriate
y_data <- data_mlca |> select(all_of(Y))

set.seed(1234)
y_data$dummy_y <- rnorm(nrow(y_data))

vif_y_model <- lm(dummy_y ~ ., data = y_data)
vif(vif_y_model)

```

public	size	awardlev	focus	research	online
1.760092	2.488298	2.215250	1.349799	2.272521	1.397997
evening	selectivity	pell	msi	nonwhite	
1.132332	1.959964	1.438918	1.491228	1.616924	

```

# Level-1 covariates (Z)
z_data <- data_mlca |>

```

```

select(all_of(Z))

set.seed(1234)
z_data$dummy_y <- rnorm(nrow(z_data))

vif_z_model <- lm(dummy_y ~ ., data = z_data)
vif(vif_z_model)

```

i_perc_white	i_perc_black	i_perc_hisp	i_perc_asian	housing	rurality
5.605996	4.024443	3.505072	1.664504	1.347178	1.113060
religious					
1.213664					

```

# Level-2 covariates (Zh)
zh_data <- data_mlca |>
  select(all_of(Zh)) |>
  mutate(cbsatype = as.factor(cbsatype)) # switch nominal variable to factor

zh_data$dummy_y <- rnorm(nrow(zh_data))

vif_zh_model <- lm(dummy_y ~ ., data = zh_data)
vif(vif_zh_model)

```

cbsatype	inst_num	tot_pop	pop_density	perc_ba
1.489369	55.185840	51.956170	14.507700	3.756909
med_income	perc_pov	perc_unemployed	perc_white	perc_black
7.506924	4.332139	2.383692	31.438482	9.009204
perc_hisp	perc_asian			
19.233788	8.332143			

```

set.seed(1234)

model_selection <- multiLCA(
  data = data_mlca,
  Y = Y,
  iT = 6:12,
  Z = Z,
  iM = 2:6,
  id_high = c("cbsa_id"),
  Zh = Zh,

```

```

incomplete      = FALSE,
extout          = TRUE,
verbose         = TRUE,
sequential      = TRUE,
reord           = FALSE
)

```

```

Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
Fitting multilevel measurement model (step 2)...
Fitting multilevel measurement model (step 2)...
Fitting multilevel measurement model (step 2)...
Fitting multilevel measurement model (step 2)...
Fitting multilevel measurement model (step 2)...
Fitting multilevel measurement model (step 3)...
Fitting multilevel measurement model (step 3)...
Fitting multilevel measurement model (step 3)...
Fitting multilevel measurement model (step 3)...
Fitting multilevel measurement model (step 3)...
Fitting multilevel measurement model (step 3)...
Fitting multilevel measurement model (step 3)...

```

```

$step1
      BIClow  BIChigh  AIC      ICL_BIClow ICL_BIChigh
iT=6  62283.9  61978.02 61019.51 -          -
iT=7  62168.61 61811.5  60692.47 -          -
iT=8  61791.57 61383.23 60103.69 -          -
iT=9  61450.78 60991.22 59551.16 -          -
iT=10 61378.01 60867.22 59266.65 -          -
iT=11 61477.86 60915.85 59154.76 -          -
iT=12 61356.19 60742.95 58821.34 -          -

```

```

$step2
      BIClow  BIChigh  AIC      ICL_BIClow ICL_BIChigh
iT*,iM=2 60880.57 60249.77 58273.13 62362.99  60480.64
iT*,iM=3 60878.13 60229.76 58198.09 62190.31  60654.52
iT*,iM=4 60810.75 60144.83 58058.12 62121.15  61154.8
iT*,iM=5 60565.22 59881.73 57739.99 61642.73  60838.63

```

iT*,iM=6 60837.8 60136.74 57939.97 62139.26 60999.12

\$step3

	BIClow	BIChigh	AIC	ICL_BIClow	ICL_BIChigh
iT=6,iM*	62424.84	62083.83	61015.25	63118.72	63193.81
iT=7,iM*	61335.85	60937.76	59690.32	62087.45	61711.67
iT=8,iM*	60911.53	60456.36	59030.06	61723.31	61243.72
iT=9,iM*	60776.01	60263.75	58658.59	61592.88	61428.86
iT=10,iM*	60706.11	60136.78	58352.76	61663.05	60758.85
iT=11,iM*	60535.22	59908.81	57945.93	61632.54	61156.46
iT=12,iM*	60564.46	59880.97	57739.23	61637.16	60857.18

\$optimal

iT= 11

iM= 5

```
analytic_model <- model_selection
```

```
analytic_model
```

CALL:

```
multiLCA(data = data_mlca, Y = Y, iT = 6:12, id_high = c("cbsa_id"),  
  iM = 2:6, Z = Z, Zh = Zh, incomplete = FALSE, extout = TRUE,  
  sequential = TRUE, reord = FALSE, verbose = TRUE)
```

SPECIFICATION:

Multilevel LC model

MISSING VALUES ON THE INDICATORS:

None

FINAL SAMPLE SIZE:

3133

ESTIMATION DETAILS:

EMiter LLfirst LLlast

1000 -29005.8 -28544.96

GROUP PROPORTIONS:

P(G1) 0.0028
P(G2) 0.1945
P(G3) 0.3613
P(G4) 0.3052
P(G5) 0.1362

CLASS PROPORTIONS:

	G1	G2	G3	G4	G5
P(C1 G)	0.0547	0.3136	0.2588	0.2493	0.0329
P(C2 G)	0.0680	0.0000	0.0620	0.0279	0.0624
P(C3 G)	0.0916	0.0000	0.1076	0.1307	0.2895
P(C4 G)	0.0625	0.0000	0.0879	0.1107	0.0455
P(C5 G)	0.0597	0.0611	0.0593	0.0001	0.0565
P(C6 G)	0.0000	0.0911	0.0716	0.1238	0.0024
P(C7 G)	0.0854	0.0000	0.0475	0.0000	0.2716
P(C8 G)	0.0146	0.5341	0.1955	0.2926	0.0000
P(C9 G)	0.3829	0.0000	0.0376	0.0649	0.0152
P(C10 G)	0.1329	0.0000	0.0192	0.0000	0.1307
P(C11 G)	0.0478	0.0000	0.0530	0.0000	0.0934

RESPONSE PROBABILITIES:

	C1	C2	C3	C4	C5	C6	C7	C8
P(public.0 C)	0.8074	0.5050	0.0988	0.8847	0.7410	0.0486	0.0061	0.0032
P(public.1 C)	0.1926	0.4950	0.0046	0.0836	0.2590	0.9514	0.9939	0.9968
P(public.2 C)	0.0000	0.0000	0.8967	0.0317	0.0000	0.0000	0.0000	0.0000
P(size.0 C)	0.1802	0.0000	0.8615	0.2961	0.6023	0.0000	0.0206	0.0364
P(size.1 C)	0.7043	0.0345	0.1337	0.6977	0.3803	0.0194	0.1830	0.6182
P(size.2 C)	0.0970	0.1984	0.0028	0.0062	0.0175	0.3295	0.3486	0.2642
P(size.3 C)	0.0125	0.2794	0.0020	0.0000	0.0000	0.4491	0.3083	0.0753
P(size.4 C)	0.0060	0.4877	0.0000	0.0000	0.0000	0.2020	0.1396	0.0059
P(awardlev.0 C)	0.0277	0.0000	0.9397	0.0288	0.3150	0.0000	0.9886	0.9680
P(awardlev.1 C)	0.4032	0.0000	0.0603	0.8829	0.5792	0.0361	0.0000	0.0234
P(awardlev.2 C)	0.2963	0.0080	0.0000	0.0871	0.0865	0.4732	0.0021	0.0060
P(awardlev.3 C)	0.2727	0.9920	0.0000	0.0012	0.0193	0.4907	0.0093	0.0026
P(focus.0 C)	0.0086	0.0000	0.0968	0.0000	0.0000	0.0000	0.0000	0.0000
P(focus.1 C)	0.1617	0.7989	0.0053	0.1258	0.1808	0.3463	0.7934	0.4447

P(focus.2 C)	0.6729	0.0970	0.0437	0.0389	0.4720	0.6464	0.1835	0.4654
P(focus.3 C)	0.0054	0.0000	0.4674	0.0000	0.0638	0.0000	0.0199	0.0612
P(focus.4 C)	0.0429	0.0060	0.0228	0.7900	0.1791	0.0073	0.0000	0.0013
P(focus.5 C)	0.0169	0.0000	0.0494	0.0091	0.0224	0.0000	0.0000	0.0016
P(focus.6 C)	0.0693	0.0108	0.3148	0.0000	0.0285	0.0000	0.0032	0.0257
P(focus.7 C)	0.0084	0.0873	0.0000	0.0362	0.0068	0.0000	0.0000	0.0000
P(focus.8 C)	0.0140	0.0000	0.0000	0.0000	0.0466	0.0000	0.0000	0.0000
P(research.0 C)	0.9540	0.0000	1.0000	0.8328	0.9143	0.2760	1.0000	1.0000
P(research.1 C)	0.0460	0.0590	0.0000	0.1672	0.0857	0.2451	0.0000	0.0000
P(research.2 C)	0.0000	0.1791	0.0000	0.0000	0.0000	0.2808	0.0000	0.0000
P(research.3 C)	0.0000	0.7619	0.0000	0.0000	0.0000	0.1980	0.0000	0.0000
P(online.0 C)	0.0768	0.1401	0.7353	0.7611	0.2122	0.0000	0.0061	0.0019
P(online.1 C)	0.5042	0.7859	0.1032	0.1829	0.3862	0.4262	0.0313	0.0591
P(online.2 C)	0.2858	0.0654	0.0632	0.0560	0.1708	0.4780	0.3792	0.4867
P(online.3 C)	0.1332	0.0085	0.0983	0.0000	0.2308	0.0958	0.5834	0.4522
P(evening C)	0.5752	0.5532	0.7012	0.0612	0.6280	0.5069	0.7867	0.6594
P(selectivity.0 C)	0.0329	0.0000	0.8514	0.0181	0.5108	0.0357	0.9962	0.9937
P(selectivity.1 C)	0.5916	0.2240	0.0950	0.3050	0.2096	0.7874	0.0038	0.0009
P(selectivity.2 C)	0.3528	0.4358	0.0383	0.4040	0.2602	0.1768	0.0000	0.0054
P(selectivity.3 C)	0.0227	0.3402	0.0153	0.2729	0.0194	0.0000	0.0000	0.0000
P(pell.0 C)	0.0048	0.0000	0.0129	0.0409	0.0073	0.0000	0.0159	0.0069
P(pell.1 C)	0.3672	0.8443	0.0578	0.6640	0.0325	0.5071	0.5824	0.4593
P(pell.2 C)	0.5783	0.1557	0.1574	0.2551	0.2546	0.4929	0.3512	0.4892
P(pell.3 C)	0.0497	0.0000	0.7719	0.0401	0.7057	0.0000	0.0505	0.0447
P(msi C)	0.0226	0.0871	0.0000	0.0352	0.8840	0.0153	0.7172	0.0598
P(nonwhite.0 C)	0.3788	0.1377	0.0866	0.3002	0.0000	0.5295	0.0000	0.3576
P(nonwhite.1 C)	0.4447	0.3083	0.0850	0.4174	0.0050	0.3915	0.0779	0.4044
P(nonwhite.2 C)	0.1704	0.4683	0.2485	0.2191	0.2052	0.0790	0.4194	0.2249
P(nonwhite.3 C)	0.0061	0.0857	0.5799	0.0633	0.7898	0.0000	0.5026	0.0131
	C9	C10	C11					
P(public.0 C)	0.9888	0.3506	0.3364					
P(public.1 C)	0.0112	0.6494	0.0000					
P(public.2 C)	0.0000	0.0000	0.6636					
P(size.0 C)	0.9804	0.0000	0.5394					
P(size.1 C)	0.0196	0.3097	0.3027					
P(size.2 C)	0.0000	0.3154	0.0820					
P(size.3 C)	0.0000	0.1874	0.0294					
P(size.4 C)	0.0000	0.1875	0.0465					
P(awardlev.0 C)	0.3523	0.0000	0.2780					
P(awardlev.1 C)	0.6477	0.1227	0.3943					
P(awardlev.2 C)	0.0000	0.4128	0.1919					
P(awardlev.3 C)	0.0000	0.4645	0.1358					
P(focus.0 C)	0.0369	0.0000	0.0000					

P(focus.1 C)	0.0310	0.5411	0.0129
P(focus.2 C)	0.0557	0.4007	0.2429
P(focus.3 C)	0.0558	0.0000	0.0000
P(focus.4 C)	0.0169	0.0342	0.0611
P(focus.5 C)	0.0068	0.0059	0.2402
P(focus.6 C)	0.0766	0.0181	0.3802
P(focus.7 C)	0.0061	0.0000	0.0064
P(focus.8 C)	0.7142	0.0000	0.0563
P(research.0 C)	1.0000	0.4295	1.0000
P(research.1 C)	0.0000	0.2379	0.0000
P(research.2 C)	0.0000	0.2231	0.0000
P(research.3 C)	0.0000	0.1095	0.0000
P(online.0 C)	0.7439	0.0182	0.0542
P(online.1 C)	0.0485	0.4873	0.2033
P(online.2 C)	0.0935	0.3639	0.1000
P(online.3 C)	0.1141	0.1306	0.6425
P(evening C)	0.1403	0.6799	0.6038
P(selectivity.0 C)	0.4348	0.0000	0.7284
P(selectivity.1 C)	0.2844	0.6636	0.1715
P(selectivity.2 C)	0.2132	0.3064	0.0581
P(selectivity.3 C)	0.0675	0.0301	0.0420
P(pell.0 C)	0.0638	0.0000	0.0288
P(pell.1 C)	0.1778	0.0539	0.0658
P(pell.2 C)	0.3543	0.5300	0.3211
P(pell.3 C)	0.4040	0.4161	0.5843
P(msi C)	0.0000	0.8414	0.0169
P(nonwhite.0 C)	0.8333	0.0000	0.0000
P(nonwhite.1 C)	0.1014	0.0132	0.0984
P(nonwhite.2 C)	0.0302	0.3732	0.4984
P(nonwhite.3 C)	0.0352	0.6136	0.4032

MODEL AND CLASSIFICATION STATISTICS:

R2entrlo	0.9224
R2entrho	0.3562
BIClo	60535.2194
BICho	59908.8095
ICLBIClo	61632.5355
ICLBICho	61156.4629
AIC	57945.928

“{r}

```
set.seed(1234)
```

testing 8 lower-level classes, 2 higher-level clusters

```
test_l8_h2<- multiLCA( data = data_mlca, Y = Y, iT = 8, Z = Z, iM = 2, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```

testing 9 lower-level classes, 2 higher-level clusters

```
test_l9_h2<- multiLCA( data = data_mlca, Y = Y, iT = 9, Z = Z, iM = 2, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```

testing 10 lower-level classes, 2 higher-level clusters

```
test_l10_h2<- multiLCA( data = data_mlca, Y = Y, iT = 10, Z = Z, iM = 2, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```

testing 11 lower-level classes, 2 higher-level clusters

```
test_l11_h2<- multiLCA( data = data_mlca, Y = Y, iT = 11, Z = Z, iM = 2, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```

testing 8 lower-level classes, 3 higher-level clusters

```
test_l8_h3<- multiLCA( data = data_mlca, Y = Y, iT = 8, Z = Z, iM = 3, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```

testing 9 lower-level classes, 3 higher-level clusters

```
test_l9_h3<- multiLCA( data = data_mlca, Y = Y, iT = 9, Z = Z, iM = 3, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```


NO SOLUTION FOUND

testing 10 lower-level classes, 3 higher-level clusters

test_l10_h3<- multiLCA(

data = data_mlca,

Y = Y,

iT = 10,

Z = Z,

iM = 3,

id_high = c("cbsa_id"),

Zh = Zh,

incomplete = FALSE,

extout = TRUE,

verbose = TRUE,

sequential = TRUE,

reord = FALSE

)

testing 11 lower-level classes, 3 higher-level clusters

test_l11_h3 <- multiLCA(data = data_mlca, Y = Y, iT = 11, Z = Z, iM = 3, id_high = c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential

```
= TRUE, reord = FALSE )
```

testing 8 lower-level classes, 4 higher-level clusters

```
test_l8_h4 <- multiLCA( data = data_mlca, Y = Y, iT = 8, Z = Z, iM = 4, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```


NO SOLUTION FOUND

testing 9 lower-level classes, 4 higher-level clusters

```
test_l9_h4<- multiLCA(  
  data = data_mlca,  
  Y = Y,  
  iT = 9,  
  Z = Z,  
  iM = 4,  
  id_high = c("cbsa_id"),  
  Zh = Zh,  
  incomplete = FALSE,  
  extout = TRUE,  
  verbose = TRUE,  
  sequential = TRUE,  
  reord = FALSE  
)
```

testing 10 lower-level classes, 4 higher-level clusters

```
test_l10_h4<- multiLCA( data = data_mlca, Y = Y, iT = 10, Z = Z, iM = 4, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential
```

= TRUE, reord = FALSE)

NO SOLUTION FOUND

testing 11 lower-level classes, 4 higher-level clusters

test_l11_h4<- multiLCA(

data = data_mlca,

Y = Y,

iT = 11,

Z = Z,

iM = 4,

id_high = c("cbsa_id"),

Zh = Zh,

incomplete = FALSE,

extout = TRUE,

verbose = TRUE,

sequential = TRUE,

reord = FALSE

)

NO SOLUTION FOUND

testing 8 lower-level classes, 5 higher-level clusters

test_l8_h5 <- multiLCA(

data = data_mlca,

Y = Y,

```

= TRUE, reord = FALSE )
test_l8_h2 test_l9_h2 test_l10_h2 test_l11_h2
test_l8_h3 test_l9_h3 #test_l10_h3 test_l11_h3
test_l8_h4 #test_l9_h4 test_l10_h4 #test_l11_h4
#test_l8_h5 #test_l9_h5 #test_l10_h5 test_l11_h5

```

```

# ``{r}

```

```

set.seed(1234)

```

```

#Single-level LCA without covariates

```

```

test_1 <- multiLCA(
  data = data_mlca,
  Y      = Y,
  iT     = 6:12,
  # Z     = Z,
  # iM    = 2:5,
  # id_high = c("cbsa_id"),
  # Zh    = Zh,
  incomplete = FALSE,
  extout     = TRUE,
  verbose    = TRUE,
  sequential = TRUE,
  reord      = FALSE
)

```

```

#Single-level LCA with covariates

```

```

test_1c <- multiLCA(
  data = data_mlca,
  Y      = Y,
  iT     = 6:12,
  Z      = Z,
  # iM    = 2:5,
  # id_high = c("cbsa_id"),
  # Zh    = Zh,
  incomplete = FALSE,
  extout     = TRUE,

```

```

    verbose      = TRUE,
    sequential    = TRUE,
    reord         = FALSE
)

```

```

test_1
test_1c

```

```

summary(test_1)
summary(test_1c)

```

```

#Full multilevel analysis with covariates

```

```

final_model <- multiLCA(
  data = data_mlca,
  Y      = Y,
  iT     = 6:12,
  Z      = Z,
  iM     = 2:5,
  id_high = c("cbsa_id"),
  Zh     = Zh,
  incomplete = FALSE,
  extout  = TRUE,
  verbose = TRUE,
  sequential = TRUE,
  reord   = FALSE
)

```

```

final_model
summary(final_model)
plot(final_model)

```

```

l1_proportions <- as.data.frame(analytic_model$mPi) # Lower-level class proportions given t
l1_proportions

```

	G1	G2	G3	G4	G5
P(C1 G)	5.465101e-02	3.136043e-01	0.25875494	2.492676e-01	3.290808e-02

```

P(C2|G) 6.797216e-02 0.000000e+00 0.06196336 2.791592e-02 6.235159e-02
P(C3|G) 9.159683e-02 4.098103e-59 0.10761461 1.307217e-01 2.895167e-01
P(C4|G) 6.245762e-02 0.000000e+00 0.08791291 1.106568e-01 4.550694e-02
P(C5|G) 5.966505e-02 6.112956e-02 0.05927720 8.514470e-05 5.645005e-02
P(C6|G) 1.608687e-75 9.113184e-02 0.07157701 1.238074e-01 2.351024e-03
P(C7|G) 8.535951e-02 0.000000e+00 0.04751956 2.582901e-24 2.716226e-01
P(C8|G) 1.463270e-02 5.341343e-01 0.19552658 2.926145e-01 2.259672e-152
P(C9|G) 3.828915e-01 0.000000e+00 0.03764605 6.493096e-02 1.521503e-02
P(C10|G) 1.329241e-01 0.000000e+00 0.01920237 0.000000e+00 1.306879e-01
P(C11|G) 4.784952e-02 0.000000e+00 0.05300543 0.000000e+00 9.339012e-02

```

```
l1_probabilities <- as.data.frame(analytic_model$mPhi)
```

```
l2_posterior <- as.data.frame(analytic_model$mPW)
```

```

l2_classes <- data.frame(
  group_id = row.names(analytic_model$mPW),
  L2_Predicted_Class = apply(l2_posterior, 1, which.max),
  L2_Prob_Class1 = l2_posterior[, 1],
  L2_Prob_Class2 = l2_posterior[, 2],
  L2_Prob_Class3 = l2_posterior[, 3]
)

```

```

l2_class_data <- data_mlca %>%
  mutate(cbsa_id = as.character(cbsa_id)) %>%
  left_join(l2_classes,
    by = c("cbsa_id" = "group_id"))

```

```

l2_class_summary <- l2_class_data %>%
  select(cbsa, L2_Predicted_Class ) %>%
  distinct()

```

```
inst_cbsa_classification <- l2_class_data
```

```
write_csv(inst_cbsa_classification, "../data_paper2/inst_cbsa_classification.csv")
```